MEKELLE UNIVERSITY

MEKELLE INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Title: Tigrigna Fake News Detection in Social Medias Using Natural Language Processing (NLP)

Ahuthors:

1. Mewcha Haftu (MIT/UR/164/11)
2. Danait G/slasse (MIT/UR/285/12)
3. Kiros Asefa (MIT/UR/189/12)
4. Haftom Etsay (MIT/UR/066/12)
5. Teklay K/mariam (MIT/UR/133/12)

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LIST OF ABBREVIATIONS

API: Application Programming Interface

AUC: Area Under Curve

BERT: Bidirectional Encoder Representations from Transformers

BiLSTM: Bidirectional Long Short-Term Memory

BOW: Bag of Words

CNN: Convolutional Neural Network

CRF: Conditional Random Field

CSV: Comma-Separated Values

FN: Fake News

GRU: Gated Recurrent Unit

IEEE: Institute of Electrical and Electronics Engineers

LIME: Local Interpretable Model-agnostic Explanations

LSTM: Long Short-Term Memory

MCC: Matthews Correlation Coefficient

ML: Machine Learning

MTL: Multi-Task Learning

NER: Named Entity Recognition

NLP: Natural Language Processing

POS: Part-of-Speech

RBF: Radial Basis Function

RNN: Recurrent Neural Network

SERA: Sera (Ethiopic Representation in ASCII)

SVM: Support Vector Machine

TF-IDF: Term Frequency-Inverse Document Frequency

TiALD: Tigrinya Abusive Language Detection

TiDG: Tigrinya Diverse Genres

XLM-R: Cross-lingual Language Model - RoBER

1. INTRODUCTION

1.1 Background

The proliferation of fake news on social media poses significant threats to information integrity and social stability worldwide. In Tigray and across Tigrigna-speaking communities in Eritrea and the diaspora, misinformation has particular urgency given recent socio-political challenges from 2020 to 2025 [1]. Natural Language Processing (NLP) offers powerful techniques for detecting misleading information by analyzing linguistic patterns in textual content [2].

Tigrigna, a Semitic language spoken by approximately 10 million people primarily in Ethiopia and Eritrea, remains severely underrepresented in NLP research [3]. Recent surveys analyzing over 50 studies from 2011 to 2025 show Tigrigna exemplifies the technological divide affecting low-resource languages, with minimal datasets and tools compared to high-resource languages [3], [4]. The language presents unique challenges including complex word morphology and the Ge'ez script with over 200 distinct symbols [5].

Recent initiatives demonstrate growing recognition of this need. UNESCO-supported fact-checking institution Haq Check monitors disinformation claims in Tigrigna and four other Ethiopian languages, debunking approximately seven stories monthly since June 2021 [6]. Sustainable Technology Solutions developed a Tigrinya AI fact-checking tool [1]. Significant research advances include the Tigrinya Diverse Genre Corpus (TiDG) with 8,067 documents [7], the Tigrinya Named Entity Corpus (TiNC24) with over 200,000 words [5], and the Tigrinya Abusive Language Dataset (TiALD) with 13,717 YouTube comments [8].

Despite these foundational resources, no study has specifically addressed fake news detection for Tigrigna, leaving a critical gap in protecting Tigrigna-speaking communities from misinformation [3], [9].

1.2 Problem Statement

Despite the critical need for misinformation detection in Tigrigna-speaking communities, no comprehensive fake news detection system exists for this language. This gap creates several interconnected challenges.

Existing fact-checking efforts by organizations like Haq Check are predominantly manual, debunking only seven stories monthly across five languages—an approach that cannot scale to match social media volume or intervene before harm occurs [6]. Supervised machine learning approaches require large labeled datasets of posts verified as "fake" or "real," but no such dataset exists for Tigrigna. Major tech platforms like Meta, X, and TikTok monitor content in only eight African languages out of over 2,000 spoken on the continent; Tigrigna is not among them, allowing harmful content to thrive undetected [10].

Tigrigna's complex morphology and Ge'ez script require dedicated preprocessing that general-purpose multilingual systems cannot provide [5], [8]. Approximately 64% of Tigrigna social media content uses Romanized transliterations rather than native Ge'ez script, adding another layer of complexity [8]. Fundamental questions remain unanswered about what linguistic features distinguish fake from real news in Tigrigna, how misinformation tactics manifest in this context, and whether techniques from related tasks like stance detection (82% accuracy) [9] or topic classification (90.23% F1-score) [7] can transfer effectively to fake news detection.

This project confronts these challenges by creating the first dedicated fake news detection system for Tigrigna, including dataset creation, model development, and prototype demonstration.

1.3 Research Questions

Primary Research Question

- How can Natural Language Processing techniques be effectively adapted to detect fake news in Tigrigna social media content?

Secondary Research Questions

- **RQ1 (Data):** What constitutes a representative and high-quality dataset for Tigrigna fake news detection, and how can such a dataset be constructed given the constraints of a low-resource language?

- **RQ2 (Features):** What linguistic, stylistic, and contextual features are most discriminative for distinguishing fake from real news in Tigrigna?
- **RQ3 (Models):** How do different modeling approaches—traditional machine learning, deep learning, and transfer learning—compare in performance for Tigrigna fake news detection?
- **RQ4 (Transfer):** To what extent can knowledge from related tasks (topic classification, sentiment analysis, abusive language detection) improve fake news detection performance through transfer learning?
- **RQ5 (Explainability):** Can model predictions be accompanied by human-interpretable explanations that build trust and enable effective human oversight?

1.4 Project Objectives

1.4.1 General Objective

To develop a culturally-aware fake news detection system for Tigrigna by creating a labeled dataset, building tailored preprocessing pipelines, and implementing comparative machine learning and deep learning models, culminating in a functional prototype for real-time prediction.

1.4.2 Specific Objectives

1. Dataset Creation: Collect 8,000 raw Tigrigna social media posts from Facebook, YouTube, and Telegram; recruit and train 3-5 native speakers as annotators; develop culturally-sensitive annotation guidelines; achieve inter-annotator agreement (Cohen's Kappa) of at least 0.75; release a labeled dataset of 5,000 items for research.

2. Preprocessing Pipeline Development: Create a pipeline tailored to Tigrigna's linguistic characteristics with robust handling for both Ge'ez and Latin scripts, evaluating SentencePiece tokenization which has shown superior performance for Tigrigna [5].

3. Model Development and Comparison: Implement baseline traditional machine learning (Logistic Regression, SVM, Random Forest); develop deep learning architectures (LSTM, Bi-LSTM, GRU with attention); apply transfer learning with pre-trained models like TiRoBERTa which achieved 90.23% F1 on topic classification [7]; create ensemble methods combining multiple approaches.

4. System Integration and Prototype Development: Build Mobile with Flutter.

5. Evaluation and Documentation: Assess all models using comprehensive metrics (accuracy, precision, recall, F1-score, AUC-ROC); thoroughly document all methodologies and results; prepare final thesis and research paper.

1.5 Scope and Limitations

The project focuses exclusively on the Tigrigna language, including both Ge'ez script and Latin-based transliterations commonly used online. Data will be collected from publicly accessible posts on Facebook, YouTube, and Telegram relevant to Ethiopian and Eritrean socio-political contexts. The core task is binary classification identifying content as "fake news" or "real news," with an optional "unverifiable" category where certainty is low. Deliverables include a labeled dataset, trained models, and a web-based demonstration prototype.

The system will process text-only content, not analyzing accompanying images or videos. The prototype will be a standalone web application without direct platform integration. While code-switching may occur, processing other languages exclusively falls outside project scope. This is a research and prototyping effort that will not maintain a live, continuously updated fact-checking database.

1.6 Significance of the Study

The project supports information integrity in Tigrigna-speaking communities vulnerable to misinformation, addressing UNESCO's identified need for countering disinformation in Ethiopian peacebuilding [6]. It establishes the first benchmark for Tigrigna fake news detection, advancing NLP research for low-resource languages that urgently need dedicated computational resources [3], [4]. It demonstrates approaches for handling morphologically rich languages with non-Latin scripts, addressing the linguistic blind spot leaving approximately 64% of Tigrigna social media content using Romanized script unchecked by current moderation systems [8], [10].

2. LITERATURE REVIEW

2.1 Theoretical Framework

This project is grounded in information processing theory, linguistic pragmatics, and computational linguistics frameworks for low-resource language NLP. Information processing theory examines how individuals receive and process information, helping explain why certain types of misinformation are more persuasive. Linguistic pragmatics studies how context contributes to meaning, which is essential for understanding how fake news exploits pragmatic features such as implicature and presupposition [2].

For low-resource languages like Tigrigna, key strategies include transfer learning from related tasks or languages, data augmentation, and careful adaptation of techniques developed for high-resource languages [3], [4].

2.2 Fake News Detection Using NLP

Automated fake news detection typically employs machine learning classifiers trained on linguistic features extracted from text [2]. Common approaches are summarized in

Common approaches include **traditional machine learning algorithms** such as Logistic Regression, Support Vector Machines (SVM), and Random Forests, which rely on manually engineered features like n-grams, readability scores, and syntactic patterns. While these models are computationally efficient and highly interpretable, they are often surpassed by **deep learning architectures** like CNNs and LSTMs. These neural models learn hierarchical representations and capture complex semantic patterns, effectively eliminating the need for manual feature engineering. Most recently, **Transformer-based models** such as BERT, RoBERTa, and Tigrigna-specific variants like **TiRoBERTa** have set new state-of-the-art benchmarks. By utilizing self-attention mechanisms and contextual embeddings, these models excel in low-resource settings through transfer learning, making them highly effective for the nuances of Tigrigna misinformation.

Recent studies have demonstrated that transformer-based models achieve state-of-the-art performance across multiple languages, with fine-tuned multilingual models like XLM-R and Afro-XLM-R showing strong results on African language tasks [7], [8].

2.3 NLP Resources for Tigrigna

Recent years have seen significant progress in developing Tigrigna NLP resources, as documented in comprehensive surveys covering over 50 studies from 2011 to 2025 [3], [4].

Existing Computational Resources

The current landscape of Tigrigna NLP research provides several foundational resources that serve as benchmarks for this study. These include:

- **TiDG (Diverse Genre Corpus):** A multifaceted corpus comprising 8,067 documents across seven categories. It is primarily used for topic classification, achieving high-performance baselines that inform our own classification targets [7].
- **TiNC24 (Named Entity Corpus):** A large-scale, human-labeled dataset containing over 200,000 words. It provides extensive annotations for Named Entity Recognition (25 tags) and Parts-of-Speech (12 tags) across ten diverse domains [5].
- **TiALD (Abusive Language Dataset):** A comprehensive multi-task benchmark consisting of 13,717 comments. It is annotated for abusiveness, sentiment, and topic, providing a critical reference for identifying harmful or inflammatory language patterns [8].
- **Stance DB (Political Stance Dataset):** A specialized collection of ,500 comments focused on political discourse. This dataset is essential for understanding stance detection, which is a significant component of identifying polarized fake news [9].
- **TiRoBERTa and TiBERT:** State-of-the-art pre-trained Transformer models specifically optimized for Tigrigna. These models have demonstrated superior performance in general text classification tasks, with TiRoBERTa reaching an F1-score of 90.2% on standard benchmarks [7].

2.4 Related Work in Tigrigna Text Classification

Political Stance Detection: A significant study by Gebrehiwot (2025) at Mekelle University investigated political stance detection in Tigrigna Facebook comments collected during the Tigray conflict period [9]. The research employed Gated Recurrent Unit (GRU), Transposed GRU, and LSTM models with Word2Vec embeddings (Skip-gram and CBOW variants). Results demonstrated that the Transposed GRU model with Skip-gram Word2Vec achieved 82% accuracy and 0.8822 F1-score, outperforming standard GRU (79%) and LSTM (77%) [9]. This study demonstrates the viability of deep learning for Tigrigna social media text analysis and establishes a benchmark for related tasks.

Topic Classification on TiDG: Gebretatios et al. (2025) evaluated multiple approaches for multi-class topic classification on the TiDG corpus [7]. Models evaluated included traditional ML (SVM with TF-IDF: 72.31% F1), Bi-LSTM with attention (78.46% F1), and transformer models (XLM-R: 85.13% F1, Afro-XLM-R: 87.41% F1, TiRoBERTa: 90.23% F1). Key findings include that monolingual pre-training (TiRoBERTa) significantly outperforms multilingual models, and SERA transliteration with SentencePiece tokenization provides optimal preprocessing.

Named Entity Recognition on TiNC24: Research on TiNC24 established strong NER performance with Transformer-based models achieving 90.18% F1-score [5]. SentencePiece tokenization with 16,000 subword units proved optimal, and SERA transliteration simplified processing without losing performance.

Abusive Language Detection on TiALD: Gaim et al. (2025) established baselines for multiple tasks on the TiALD dataset [8]. Results include abusive language detection (81.4% macro F1), sentiment analysis (73.2% macro F1), and topic classification (78.1% macro F1). Key findings indicate that models perform better on Ge'ez script comments than Latin script, and multi-task learning shows promise for improving individual task performance.

2.5 Research Gap Analysis

Despite the significant advances in Tigrigna NLP resources and demonstrated success of various text classification approaches, several critical gaps remain:

Task-Specific Gap: No existing research has specifically addressed fake news detection for Tigrigna [3], [4]. While related tasks (topic classification, stance detection, abusive language detection) provide useful foundations, fake news detection presents unique challenges including nuanced distinction between truth and falsehood, temporal dynamics of evolving misinformation tactics, and cultural specificity requiring localized understanding.

Dataset Gap: There is no publicly available dataset of Tigrigna social media posts annotated specifically for fake news. Existing resources focus on topic [7], named entities [5], or abusive content [8], but not misinformation detection.

Model Evaluation Gap: While pre-trained models have been evaluated on topic classification (90.23% F1) [7] and NER (90.18% F1) [5], their performance on fake news detection specifically is unknown. Different tasks may benefit from different architectural choices or preprocessing strategies.

Explainability Gap: Existing Tigrigna NLP research has focused on prediction accuracy without addressing model interpretability. For a sensitive application like fake news detection, explainability is crucial for building trust and enabling human oversight [2].

Script Handling Gap: Approximately 64% of Tigrigna social media content uses Romanized transliterations [8], but existing resources and models show performance degradation on Latin script content. This project specifically addresses this gap by ensuring robust handling of both scripts.

This project directly addresses these gaps by creating the first dedicated fake news detection system for Tigrigna, including dataset creation, systematic model comparison, explainability integration, and robust script handling.

3. METHODOLOGY

3.1 Research Design

This project adopts a Design Science Research approach, executed in seven phases to ensure a robust transition from data collection to a functional prototype:

Phase 1: Foundation (Wk 1-2): Literature review and finalization of veracity definitions.
Phase 2: Dataset Construction (Wk 3-5): Multi-platform collection and double-blind annotation.
Phase 3: Preprocessing (Wk 6-7): Cleaning, SERA normalization, and subword tokenization.
Phase 4: Model Development (Wk 8-10): Comparative implementation of ML, Bi-LSTM, and Transformers.
Phase 5: Evaluation & Explainability (Wk 11-12): Statistical testing and LIME integration.
Phase 6: System Integration (Wk 13): Building the responsive Streamlit/API architecture.
Phase 7: Documentation (Wk 14): Final thesis submission and code release.

3.2 Resources Required

3.2.1 Software and Tools

The project will primarily use Python 3.9+ for its extensive NLP ecosystem. Core machine learning development will rely on PyTorch and Hugging Face Transformers for fine-tuning TiRoBERTa, while scikit-learn will provide the baseline ML implementations. Data manipulation and text cleaning will be handled via Pandas and NLTK. For the prototype, Streamlit will be used to build a responsive, user-friendly interface. Version control and collaboration will be managed through Git and GitHub.

3.2.2 Hardware Requirements

High-performance computing is essential for training Transformer models. Development will occur on local machines with at least 16GB RAM for data preprocessing. For heavy model training and fine-tuning, the project will utilize GPU acceleration through Google Collab Pro or Kaggle Kernels to ensure efficient iteration cycles.

3.3 Data Collection Strategy

To capture the linguistic diversity of Tigrigna misinformation, data will be aggregated from the following platforms:

- Facebook: Public news pages and community groups using official Graph API/Scraping.
- Telegram: High-traffic Tigrigna news channels via the Telethon library.
- YouTube: Transcripts and comments from prominent news/commentary channels [8].
- TikTok: Trending text-overlays and descriptions (where accessible).
- X: public Posts, comments, Politicians and prominent pages
- Main Streaming Media posts for such as BBC Tigrigna, VOA and other medias

Target Size: 8,000 raw posts will be collected to produce a high-quality, filtered dataset of 3,000–5,000 annotated items.

3.4 Data Analysis

3.4.1 Data Annotation Framework

To ensure academic validity, the annotation process will be strictly regulated:

- Annotation Team: 3–5 native speakers, including at least one individual with a journalism or legal background.
- Labels: 1. Credible: Verifiable facts from trusted outlets (BBC, VOA, official bulletins).

2. Non-Credible: Fabricated claims or misleading content.
3. Unverifiable/Neutral: Subjective opinions or content lacking a factual claim.

- Quality Assurance: Target a Cohen's Kappa $\ge$ 0.75. Disagreements will be resolved through a third-party "Adjudicator" (the lead researcher).

3.4.2 Preprocessing Pipeline

The pipeline is designed to bridge the gap between Ge'ez script and Latin transliteration:

- Normalization: Unified Unicode (NFC) for Ge'ez and SERA transliteration for Latin script.
- Tokenization: SentencePiece subword tokenization (16k units) to handle Tigrigna's complex morphology [5].
- Code-Switching: Token-level identification for mixed Tigrigna-English or Tigrigna-Amharic content.

3.4.3 Feature Engineering

- Linguistic: Character (3-6) and Word n-grams with TF-IDF weighting.
- Stylistic: Punctuation density (e.g., excessive use of "!!!"), capitalization patterns, and lexical diversity.
- Contextual: Platform type, presence of URLs, and post-engagement metrics.

3.4.4 Model Development

Three distinct approaches will be compared to determine the optimal architecture:

1. Baseline (Traditional ML): Logistic Regression and SVM using TF-IDF.
2. Deep Learning: Bi-LSTM with Attention mechanism to capture long-range dependencies in Tigrigna sentences.
3. Transfer Learning: Fine-tuning TiRoBERTa (monolingual) and Afro-XLM-R (multilingual) for veracity classification.

3.4.5 Evaluation & Explainability

- Metrics: Macro F1-score (primary), AUC-ROC, and Confusion Matrices.
- Statistical Test: McNemar's Test ($\alpha = 0.05$) to confirm if model performance differences are significant.

- Explainability: Integration of LIME (Local Interpretable Model-agnostic Explanations) to highlight specific Tigrigna words that triggered a "Non-Credible" classification.

3.5 System Architecture

The final system will be a modular, layered architecture:

1. Presentation Layer: A responsive Streamlit interface for user input.
2. Service Layer: A Python-based API that orchestrates preprocessing and model inference.
3. Model Layer: The ensemble of saved weights (TiRoBERTa + Bi-LSTM).
4. Explanation Module: Generates visual heatmaps for the user to understand the prediction.

Technology Stack: Python 3.9x+, PyTorch/HuggingFace, Scikit-learn, and Streamlit.

3.6 Limitations and Challenges

3.6.1 Anticipated Challenges

The primary technical challenge is data scarcity, as there are currently no veracity-labeled datasets for Tigrigna, requiring a completely manual verification process. Ensuring high annotation quality is equally difficult because misinformation often involves subjective nuances that can lead to low agreement among human evaluators. Furthermore, the project must address class imbalance, as credible news typically outnumbers fabricated content in the wild, which can bias the model toward the majority class. From a linguistic perspective, Tigrigna's complex morphology and the dual use of Ge'ez and Latin scripts complicate the preprocessing phase. Finally, the rapid evolution of misinformation tactics poses a threat to long-term model effectiveness, as detection patterns valid today may become outdated as disinformation strategies change.

3.6.2 Mitigation Strategies

To address data scarcity and script complexity, the project will utilize transfer learning from established Tigrigna tasks such as topic classification and stance detection, leveraging models like TiRoBERTa. Annotation efficiency will be improved through an active learning approach, where the model identifies and prioritizes the most informative samples for human review, thereby maximizing the value of the 14-week timeline. To combat class imbalance, data augmentation techniques like back-translation and synonym replacement will be used to synthesize additional training examples for the minority class. The issue of evolving tactics will be mitigated by designing a modular system architecture that allows for

regular retraining with fresh data. Finally, all inherent model biases and linguistic limitations will be transparently documented in the final report to ensure academic honesty and provide a roadmap for future research.

3.7 CONCLUSION

This proposal outlines a comprehensive final year project to develop the first dedicated fake news detection system for the Tigrigna language using Natural Language Processing. The project addresses a critical gap in computational resources for a language spoken by approximately 10 million people, with particular urgency given recent socio-political challenges and the lack of content moderation for Tigrigna on major social media platforms [10].

The project builds upon recent advances in Tigrigna NLP, including the TiDG corpus (8,067 documents), TiNC24 (200K+ words), TiALD (13,717 comments), and pre-trained models like TiRoBERTa achieving 90.23% F1-score [7], [5], [8]. By adapting these resources to the novel task of fake news detection and creating the first annotated dataset specifically for this purpose, the project makes a significant contribution to both academic research and practical applications.

The methodology is designed to be feasible for undergraduate implementation while maintaining academic rigor. A 14-week timeline with clearly defined milestones, realistic targets (5,000 annotated items, ≥85% accuracy), and appropriate scope ensures successful completion. The project addresses the specific challenge of handling both Ge'ez and Latin scripts, which affects approximately 64% of Tigrigna social media content [8].

Success will contribute to:

- Academic knowledge: Good benchmark for Tigrigna fake news detection with more datasets
- Practical tools: Publicly available dataset and models
- Social impact: Foundational resource for fact-checking organizations [6]

With a clear plan, appropriate resources, and strong ethical foundations, this project is well-positioned for successful completion and meaningful contribution to low-resource language NLP.

4 EXPECTED OUTCOMES AND DELIVERABLES

4.1 Deliverables

Annotated Dataset: A labeled Tigrigna misinformation dataset containing 3,000 to 5,000 high quality samples. This will be the first of its kind for the language and will be released for research purposes.
Trained Models: A suite of optimized models including a Baseline ML, a Bi-LSTM with Attention, and a fine-tuned TiRoBERTa model, all with documented performance logs.

Source Code: A complete and reproducible GitHub repository containing the Python scripts for data scraping, Ge'ez and Latin script preprocessing, model training, and evaluation.

Web Application: A functional prototype developed using Streamlit that provides real-time credibility analysis and explainable visual feedback for user-provided Tigrigna text.///Project **Report:** A comprehensive undergraduate thesis documenting the research journey, data collection hurdles, linguistic analysis, and experimental results.
Research Paper: A formal paper summarizing the results, targeted for submission to local or international NLP workshops like AfricaNLP or similar computational linguistics venues.

4.2 Performance Targets

Given that veracity detection is more complex than standard topic classification, the target metrics are set realistically to ensure academic integrity:
Accuracy: Target of 80 percent or higher. This is ambitious yet grounded in recent Tigrigna stance detection benchmarks.

F1-score: Target of 0.82 or higher. We prioritize the macro-average F1-score to ensure the model does not ignore the minority class in the dataset.
Precision and Recall: Balanced above 0.80. This ensures we minimize "False Positives" which could wrongly label credible news as fake.

AUC-ROC: Target of 0.85 or higher to prove the model has a strong ability to distinguish between credible and non-credible content across different thresholds.

Inference Latency: Less than 1.5 seconds per request. This ensures the final prototype is responsive enough for live demonstrations during the thesis defense.

4.3 Contribution to Knowledge

Theoretical: This study provides the first systematic analysis of linguistic patterns used in Tigrigna misinformation, identifying specific stylistic markers unique to the socio-political context.
Methodological: It establishes a reproducible framework for creating veracity datasets in low-resource languages, specifically addressing the challenge of mixed Ge'ez and Latin scripts.

5. PROJECT TIMELINE

Table 5.1: Timeline of the project

WBS	**Project Initiation**	**Weeks**	**Outcomes/Deliverables**
1.0	Project Initiation	1-2	Project Charter & Literature Review
1.1	Lit. review, Ethics, & Team Recruitment	1-2	Lit. review, Ethics, & Team Recruitment
2.0	Data Collection	2-3	Raw Dataset & Extraction Tools
2.1	Tool development & Source identification	2	Scrapers/API scripts; Target source list.
2.2	Data extraction	3	Preliminary raw data repository.
3.0	Annotations	3	Gold Standard Annotated Dataset
3.1	Guidelines & Annotator Training	4	Annotation manual; Trained team.
3.2	Pilot & Main Annotation	4	Finalized high-agreement labels.
3.3	Adjudication	5-6	Processed Data Pipeline
4.0	Preprocessing	6	Processed Data Pipeline
4.1	Pipeline development & Tokenization	7	Cleaned text; Normalized Ge'ez tokens.
4.2	Feature extraction	7	Numerical feature vectors (TF-IDF/Embeddings).
5.0	Model Development	8	Trained Model Artifacts
5.1	Traditional ML & Deep Learning	8-9	Baseline scores; CNN/LSTM weights.
5.2	Transfer Learning & Ensembles	10	Optimized TiRoBERTa/Hybrid models.
6.0	Evaluation	11-12	Performance Report
6.1	Evaluation & Error Analysis	11	Confusion matrices; Error case study.
6.2	Explainability (LIME/Attention)	12	Visualized attention weights for interpretability.
7.0	System Integration	13	Functional Prototype
7.1.	Web Interface & System Testing	13	Mobile app with Flutter
8.0	Documentation	14	Final Thesis & Code Release
8.1	Thesis writing & Code release	14	Final PDF; Public GitHub repository.

6 BUDGET

While this project does not require direct financial funding for hardware procurement, it relies on specific technical infrastructure provided by the Mekelle Institute of Technology (MIT) campus and existing cloud services.

7. ETHICAL CONSIDERATIONS

This study prioritizes data privacy by collecting only publicly available social media content while anonymizing all user handles and identifying information. We will strictly comply with platform terms of service and store all data using secure, encrypted systems. No raw identifiable data will be released publicly—only aggregated, anonymized datasets will be shared for research purposes.

All annotators will provide informed consent and receive fair compensation where funding permits. Content warnings will be provided for potentially disturbing material, with annotators having the option to skip any item without penalty. Regular check-ins and support access will be maintained throughout the process.
We acknowledge that the dataset may contain misleading or sensitive content by nature. Usage will be strictly limited to academic research, with content warnings included in any public demonstrations. Native Tigrinya speakers will be consulted throughout to ensure cultural and linguistic sensitivity.

Any deployed system will clearly communicate its limitations and be framed as assistive rather than authoritative. It will not be used as an autonomous moderation tool without human oversight, and we explicitly prohibit any use for surveillance, censorship, or suppression of free expression.

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APPENDICES

Appendix A: Sample Annotation Guidelines

Definition: Fake news is fabricated information that mimics news media content in form but not in organizational process or intent, created with intention to deceive.

Categories:

- Fake News: Meets definition above
- Real News: Verifiable factual information from credible sources
- Unverifiable: Cannot determine truth value
- Not News: Opinion, personal update, advertisement

Appendix B: Data Collection Sources (Preliminary)

Facebook: Targeting VOA Tigrigna, BBC Tigrigna, and popular community pages.

YouTube: Focusing on Tigrigna news channels and high-engagement commentary channels.

Telegram: Extracting data from public news distribution and community alert channels.